

Bachelor in Computer Applications

Course Title: **Operations Research**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Solve the given Linear Programming Problem (LPP) by using simplex method and interpret the results. Minimize $Z = 2A + 10B$ Subject to constraints: $A + B = 50$, $A \geq 20$, $B \leq 40$ Where, $A, B \geq 0$
2. Find transport schedule to minimize the transportation cost for the following transportation problem. The transportation cost per unit and units demanded and available are given in the table.

	From	To	A	B	C	Units demanded
X	9	10	5	Y	10	14
8	2	0	Z	13	10	8
2	0	Units available	20	15	10	45
3. The table give below give the information about the activities, their predecessors and time duration required to complete the activities of the project.

Activity	A	B	C	D	E	F	G	H	P
Predecessor	--	A	B	C	E	F	G		
Time in weeks	5	1	2	6	3	2	6	1	4

Draw the network diagram and identify the critical activities and critical path. Also find the minimum time duration required to complete the project.
4. Callennoids Metropolitan is putting up bids for four used motorbikes company. The Metropolitan allows individuals to make bids on all four motorbikes company but will accept only one bid per individual have made the following bids (in thousands Rs).

Motorbike Company	Individuals	Honda	Hero	Bajaj	Yamaha
A	100	90	1	10	90
B	110	100	95	90	105
D	115	100	95	100	

Make the use of Hungarian method to assign the individuals to different motorbike company in order to maximize the revenue.
5. A milk salesman estimates the probability of the demand for a litre of milk is as follows:

Demand	1	11	2	13	14	15
Probability	0.1	0.0	0.1	0.5	0.3	0.2

He purchase a litre of milk @ of Rs. 60 and sells it @ of Rs. 70. Prepare payoff table and find the optimal act based on using EMV and EOL criterion determining the unsold milk has value.
6. On the average 96 patients per 24 hours day require the emergency service in clinic. Also on the average, a patient requires 10 minutes of active attention. Assume that the facility can handle only one emergency at a time. If this situation satisfy the all the conditions for apply queuing theory, find the average (expected) queue length and the waiting time for the patient to be served.
7. Determine the best strategy for each player A and B and value of the game.

Player A's strategy \ Player B's strategy	B	B	B	A
4	0	4	0	4
0	A <td>3</td> <td>0</td> <td>2</td>	3	0	2
5	0	A <td>1</td> <td>3</td>	1	3
0	2			
8. The following tables gives the three kinds of foods and three kinds of vitamin contained on them. Formulate objective function and constraint of LPP for minimizing the cost.

Vitamin	Food	Daily Requirements
F	F	F
V	20	10
10	10	10
20	10	20
40	20	10

Cost per unit of food Rs. 20 Rs. 24 Rs. 18

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9. Describe modified distribution (MODI) method used for testing the optimality of initial solution of transport problem.
 10. What is called a queue? Describe the operating characteristics of the single channel queuing model.
 11. The following activities must be completed in order to complete the project. Draw network diagram and establish relationship between the activities of the project. Activity PQRSTUVWX Predecessor--P, Q, P, R, T, U, S, UV, W
 12. Write short note on:
a. Scopes of operations research
b. Dominance Rule Method in game theory